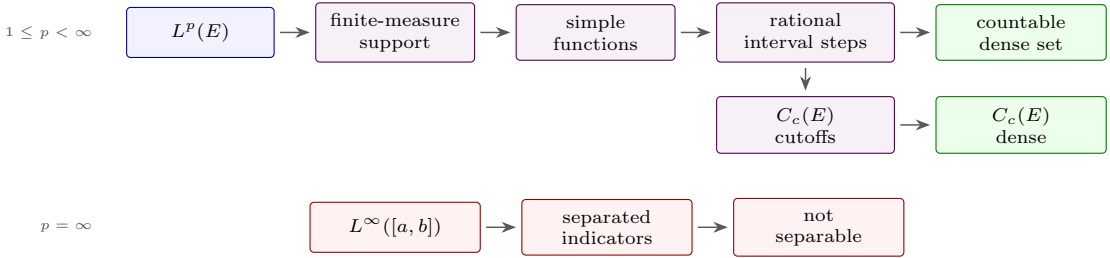
$1 \le p < \infty$
$L^p(E)$
finite-measure support
simple functions
rational interval steps
countable dense set
$C_c(E)$ cutoffs
$C_c(E)$ dense
$p = \infty$
$L^\infty([a,b])$
separated indicators
not separable